

VELORA: COMPLETE MASTER ROADMAP (v0.1 – v3.0)

Author: CTO & Principal Systems Architect
Classification: Enterprise Engineering Blueprint
Target Architecture: Next.js 14 (App Router) | PHP 8.3 REST API | MySQL 8.0 | MetaApi Bridge | Linux cPanel Infrastructure

1. Executive Roadmap Timeline

2026 Q1 - Q2	2026 Q3 - Q4	2027 Q1 - Q2
2027 Q3 - 2028 Q1		
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PHASE 1: MVP FOUNDATION	PHASE 2: CORE AUTOMATION	PHASE 3: ADVANCED AI &
PHASE 4: ENTERPRISE &		
	& PERFORMANCE PLATFORM	CROSS-PLATFORM MOAT
ECOSYSTEM SCALING		
v0.1: Core Data Engine	v0.5: Analytics Engine	v1.5: Advanced
Multi-Acc	v2.5: White-Label & B2B	
v0.2: MetaApi Cloud Bridge	v1.0: Commercial Launch	v2.0: Native EAs &
Apps	v3.0: Autonomous Trading	
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2. Comprehensive Master Roadmap Table

Version	Phase	Primary Objective	Key Deliverables	Complexity	Priority	Est. Duration
v0.1	Phase 1	Foundation & Manual	Auth, Manual Trade	Medium	P0 (Critical)	6 Weeks

Version	Phase	Primary Objective	Key Deliverables	Complexity	Priority	Est. Duration
		Schema	Engine, Basic Metrics, CRUD DB			
v0.2	Phase 1	Automated Cloud Sync Bridge	MetaApi Sync Pipeline, Webhooks, Live Balance Sync	High	P0 (Critical)	6 Weeks
v0.5	Phase 2	Analytics & Execution Profiler	Interactive Charts, R-Multiple Breakdown, Custom Tags	Medium	P1 (High)	8 Weeks
v1.0	Phase 2	Commercial Launch & AI v1	Stripe Subscriptions, AI Coach v1, RBAC Admin Panel	High	P0 (Critical)	8 Weeks
v1.5	Phase 3	Multi-Account &	Portfolio Aggregati	Medium	P1 (High)	8 Weeks

Version	Phase	Primary Objective	Key Deliverables	Complexity	Priority	Est. Duration
		Portfolio	on, Currency Conversion, Prop DD			
v2.0	Phase 3	Zero-Cost Sync & Mobile Apps	Custom MT4/MT5 EAs, React Native iOS/Android Apps	Very High	P1 (High)	12 Weeks
v2.5	Phase 4	B2B, Social & Copy Trading	Prop Firm White-Label, Verification Seals, Copy Engine	Very High	P2 (Medium)	12 Weeks
v3.0	Phase 4	Sovereign AI Trading OS	Voice AI Assistant, ML Predictive Win Models, Open API	Extremely High	P2 (Medium)	16 Weeks

3. CTO Strategic Recommendations & Architectural Pivots

- **MetaApi Cost & Reliability Failover Strategy:** MetaApi costs grow linearly with connected accounts. *Pivot:* Use MetaApi for initial onboarding, but aggressively push users to install our free **Custom Native MetaTrader EAs (v2.0)**. The EA transmits trades directly via encrypted HTTPS POST to our API, bypassing MetaApi entirely, reducing operating overhead to near \$0 per user while dropping sync latency to under 50ms.
- **Decoupled Analytics Engine:** Never compute complex financial stats (Sharpe Ratio, Expectancy, Max Drawdown) on the fly during API request execution. *Pivot:* Implement asynchronous event-driven calculations. When a trade completes, push an event to a queue worker. A background worker recalculates metrics and updates pre-aggregated summary tables (user_analytics_daily, account_performance_summary). The UI fetches pre-calculated snapshots instantaneously.
- **Strict Client-Side Encryption for Broker Secrets:** Investor/Trading passwords must never be stored in plain text. Encrypt all bridge credentials at the field level using **AES-256-GCM** before writing to MySQL.
- **Sub-200ms Latency SLA on cPanel:** MySQL on cPanel can bottleneck on write spikes. Enforce composite indexing on (account_id, open_time) and (account_id, symbol), strict page/query caching, and lightweight JSON response payloads.

4. Phase & Version Breakdown

PHASE 1: MVP FOUNDATION & AUTOMATED BRIDGE

Version 0.1 — Core Data Engine & Manual Journal

- **Version Goal:** Establish core system architecture, authentication, database schema, and manual journaling functionality.
- **Why This Version Exists:** To create a stable, production-ready baseline layer for user authentication, trade entity storage, and initial UI layout before introducing complex async background synchronization.
- **Features:** User Registration/Login, Password Reset, JWT Handling, Manual Trade Entry Modal, Basic Trade Table, Filter by Date/Symbol, Basic Calculations (Win Rate, Total PnL, Profit Factor).
- **UI/UX Changes:** Sleek Next.js dark-mode layout with Tailwind CSS, Responsive Table with column visibility toggles, Manual Trade Modal with multi-step inputs (Entry, Exit, Lot, Commission, Swap, Stop Loss, Take Profit).
- **Backend Changes:** PHP 8.3 Layered Architecture setup (Controllers, Services, Repositories). Auth endpoints (/api/v1/auth/register, /login, /me), Trade endpoints (/api/v1/trades GET/POST/PUT/DELETE).
- **Database Changes:** Setup users, user_sessions, trading_accounts, trades, and trade_exits

tables with InnoDB engines, foreign key constraints, and 3NF normalization.

- **API Changes:** Standardized JSON REST API contract (status, data, error, timestamp). JWT Bearer Auth Headers.
- **AI Changes:** None (Infrastructure preparation only).
- **Security Changes:** Password hashing using Bcrypt (cost factor 12), strict CORS headers, SQL Injection protection via PDO prepared statements.
- **Testing Requirements:** PHPUnit tests for core mathematical service calculations; Jest unit tests for Next.js form state validation.
- **Acceptance Criteria:**
 1. A user can register, log in, and persist session across page reloads via dual JWT tokens.
 2. Manual creation of a trade accurately calculates Net PnL and updates the global summary dashboard.
 3. REST API response time for trade fetch is < 100ms for 500 rows.
- **Dependencies:** Base repository initialization, Next.js App Router boilerplate, MySQL Database provisioning.
- **Estimated Complexity:** Medium.
- **Development Priority:** P0 (Critical).

Version 0.2 — MetaApi Cloud Bridge & Real-Time Sync

- **Version Goal:** Seamless zero-manual-input trade auto-synchronization for MetaTrader 4 and 5 via MetaApi.
- **Why This Version Exists:** Manual trade logging is the #1 point of friction causing user drop-off in trading journal SaaS products. Automation is mandatory for competitive parity.
- **Features:** MetaApi Account Connection Wizard, Automated Historical Trade Sync (up to 12 months), Webhook Ingestion Engine for live trade entry/exit/modification, Connection Status Monitor.
- **UI/UX Changes:** Connection modal for MetaTrader platform selection (MT4/MT5), server search autocomplete, credential input fields, real-time sync progress bar.
- **Backend Changes:** MetaApi SDK/HTTP integration, Async Webhook Processing Controller (/api/v1/webhooks/metaapi), Background Queue worker for batch historical synchronization.
- **Database Changes:** Add metaapi_account_id, sync_status, last_synced_at, connection_credentials_encrypted fields to trading_accounts table.
- **API Changes:** New endpoints: /api/v1/accounts/connect-metaapi, /api/v1/accounts/{id}/sync, /api/v1/webhooks/metaapi.
- **AI Changes:** None.
- **Security Changes:** AES-256-GCM encryption for MetaTrader account passwords stored in DB; HMAC verification on incoming MetaApi webhook payloads.
- **Testing Requirements:** Mock MetaApi Webhook payload test suite; Sync failure retry logic

tests; Account disconnect cleanup tests.

- **Acceptance Criteria:**
 1. User inputs MT4/MT5 server, login, and investor password; account connects successfully.
 2. Historical trade logs are fully synced and accurately populated in MySQL in < 15 seconds for 1,000 trades.
 3. Live trades executed on MT4/MT5 sync to Velora within < 3 seconds via webhooks.
- **Dependencies:** Version 0.1 core schema and trade service.
- **Estimated Complexity:** High.
- **Development Priority:** P0 (Critical).

PHASE 2: CORE PLATFORM, ANALYTICS & COMMERCIAL LAUNCH

Version 0.5 — Advanced Analytics Engine & Tagging

- **Version Goal:** Deliver market-leading visual analytics, execution charts, and qualitative trade tagging.
- **Why This Version Exists:** Raw trade logs provide little value without actionable statistical visualization. This version gives traders deep insight into their execution mechanics.
- **Features:** Interactive TradingView/Recharts Equity Curve, Win Rate vs Risk-Reward Heatmap, R-Multiple Distribution Chart, Time-of-Day/Day-of-Week Profitability Matrix, Setup Tags (e.g., "Fair Value Gap"), Mistake Tags (e.g., "FOMO"), Trade Notes with Screenshot attachments.
- **UI/UX Changes:** Redesigned Analytics Dashboard tab, Tag Creation Modal, Drag-and-drop Image Upload for trade charts, Interactive Filtering drawer (Filter by Tag, Asset, Time range, Risk profile).
- **Backend Changes:** Analytics Calculation Engine (MetricsService.php), Image upload handling service with auto-compression (JPEG conversion + resize), Tag association services.
- **Database Changes:** Create tags, trade_tags, trade_attachments, and user_analytics_daily tables.
- **API Changes:** /api/v1/analytics/summary, /api/v1/analytics/equity-curve, /api/v1/analytics/heatmaps, /api/v1/trades/{id}/tags.
- **AI Changes:** Data structuring layer for future LLM ingestion.
- **Security Changes:** File upload validation (MIME-type whitelist: JPG/PNG/WebP, max file size 5MB, random file path hashing to prevent arbitrary execution).
- **Testing Requirements:** Precision verification of mathematical formulas against baseline financial test sets; Screenshot upload unit tests.
- **Acceptance Criteria:**

1. Equity curve dynamically re-renders in < 100ms when toggling filters.
 2. R-Multiple correctly handles trades without defined Stop Losses by using initial price delta fallbacks.
 3. Tag performance matrix accurately breaks down profit factor per setup.
- **Dependencies:** Version 0.2 synced trade database.
 - **Estimated Complexity:** Medium.
 - **Development Priority:** P1 (High).

Version 1.0 — Commercial Launch, AI Coach v1 & Billing

- **Version Goal:** Complete monetization infrastructure, introduce AI Behavioral Guidance, and launch commercially.
- **Why This Version Exists:** Converts Velora into a revenue-generating SaaS product while differentiating it from competitors via automated AI coaching.
- **Features:** Stripe Subscription Gateway Integration (Monthly \$29/mo, Annual \$240/yr), Tiered Access Rules, AI Coach v1 (Automated trade leak detection: over-trading, revenge trading, rule violations), Admin Management Panel, Public Verified Profile URLs.
- **UI/UX Changes:** Monetization Paywall Modals, Subscription Billing Portal in User Settings, AI Coach Insight Banners on Dashboard, RBAC Admin Console layout.
- **Backend Changes:** Stripe Webhook handling (/api/v1/webhooks/stripe), Subscription middleware checking user roles, AI Coaching Engine (AICoachService.php) utilizing OpenAI API with structured JSON outputs.
- **Database Changes:** Create subscriptions, ai_coaching_logs, and admin_audit_logs tables. Add role column to users.
- **API Changes:** /api/v1/subscriptions/checkout, /api/v1/subscriptions/portal, /api/v1/ai-coach/latest-insights, /api/v1/admin/*.
- **AI Changes:** AI Engine v1: Ingests last 50 trades JSON payload + aggregate metrics, runs prompt templates evaluating psychological bias, outputs structured JSON recommendations.
- **Security Changes:** Stripe webhook cryptographic signature verification; Strict RBAC middleware preventing standard users from reaching /admin/* routes.
- **Testing Requirements:** Full Stripe checkout workflow automated tests using test card numbers; Subscription expiration and access revocation unit tests.
- **Acceptance Criteria:**
 1. User can upgrade to Pro via Stripe Checkout and instantly unlock Pro features.
 2. Failed monthly subscription automatically downgrades account access after 3 grace-period retries.
 3. AI Coach correctly identifies revenge trading behavior (3+ consecutive losses followed by 2x lot size expansion within 15 minutes).

- **Dependencies:** Version 0.5 Analytics engine.
- **Estimated Complexity:** High.
- **Development Priority:** P0 (Critical).

PHASE 3: PROFESSIONAL PLATFORM & CROSS-PLATFORM MOAT

Version 1.5 — Multi-Account Portfolio & Prop Firm Suite

- **Version Goal:** Unified portfolio tracking, currency conversion, and specific Prop Firm drawdown protection tools.
- **Why This Version Exists:** Professional and Prop Firm traders manage 3 to 10+ evaluation accounts simultaneously across different base currencies and brokers. Solving this multi-account chaos creates a massive retention moat.
- **Features:** Multi-Account Portfolio Grouping, Auto Currency Normalization (EUR/GBP/JPY trades auto-converted to USD base currency using daily FX rates), Prop Firm Rule Monitor (Tracks Max Daily Drawdown, Max Total Loss, and Target Profit limits in real-time), Risk Exposure Dashboard.
- **UI/UX Changes:** Multi-account selector dropdown with multi-select checkboxes, Aggregate Portfolio View, Real-Time Prop Drawdown Gauge Progress Bars with customizable risk threshold warnings.
- **Backend Changes:** Currency Rate Exchange Service (Daily background job pulling ECB rate data), Multi-Account Calculation Aggregator, Real-time Drawdown Rule Evaluator.
- **Database Changes:** Create currency_rates, account_groups, prop_firm_rules tables.
- **API Changes:** /api/v1/portfolio/summary, /api/v1/account-groups, /api/v1/prop-rules/{account_id}.
- **AI Changes:** AI Coach updated to analyze cross-account correlation risk (e.g., user buying EURUSD on Account A while selling DXY on Account B).
- **Security Changes:** Account ownership strict verification in SQL queries ensuring users can only group accounts they explicitly own.
- **Testing Requirements:** Currency conversion precision tests; Prop Firm drawdown edge-case calculations (high equity peak vs balance tracking).
- **Acceptance Criteria:**
 1. Combining a EUR 10,000 account and a USD 10,000 account correctly displays normalized USD portfolio equity using live rates.
 2. Reaching 80% of daily drawdown limit triggers immediate In-App, Email, and Webhook alerts.
- **Dependencies:** Version 1.0 architecture.
- **Estimated Complexity:** Medium.
- **Development Priority:** P1 (High).

Version 2.0 — Custom Native MT4/MT5 EAs & Mobile Apps

- **Version Goal:** Deploy native MQL4/MQL5 Expert Advisors for zero-cost sync and launch native iOS/Android mobile applications.
- **Why This Version Exists:** Completely eliminates third-party MetaApi dependency costs for power users, reduces sync latency to sub-50ms, and enables mobile trade logging/tagging.
- **Features:** Custom Velora MT4 & MT5 Expert Advisors (EAs) for direct server sync, Native iOS and Android Mobile Apps (React Native), Mobile Push Notifications, Mobile Quick-Tagging interface.
- **UI/UX Changes:** React Native mobile dashboard optimized for touch navigation; EA settings installer UI on desktop web; Push notification preferences screen.
- **Backend Changes:** Direct EA Ingestion Gateway (/api/v1/ea/sync) handling lightweight HTTPS POST payloads from MetaTrader terminals; Firebase Cloud Messaging (FCM) push notification integration.
- **Database Changes:** Add ea_api_key_hash to trading_accounts. Create device_tokens table for mobile push notifications.
- **API Changes:** /api/v1/ea/handshake, /api/v1/ea/trade-event, /api/v1/devices/register-push.
- **AI Changes:** Mobile push delivery of AI Coach daily morning strategy context.
- **Security Changes:** EA Authentication via single-use rotated API secret keys; HMAC signature validation on EA transmission headers.
- **Testing Requirements:** MQL4/MQL5 stress tests across 100 simultaneous simulated MetaTrader terminals; Mobile app automated build tests via Fastlane.
- **Acceptance Criteria:**
 1. EA installed on MT4/MT5 transmits trade closing event to Velora DB within < 50ms.
 2. Mobile app displays real-time push notification upon trade closure allowing one-tap tagging.
 3. Zero third-party API charges incurred for EA-synced accounts.
- **Dependencies:** Version 1.5 portfolio APIs.
- **Estimated Complexity:** Very High.
- **Development Priority:** P1 (High).

PHASE 4: ENTERPRISE, B2B & SOVEREIGN AI ECOSYSTEM

Version 2.5 — B2B White-Labeling, Social Verification & Copy Trading

- **Version Goal:** Expand into B2B Prop Firm white-labeling, verified trader social profiles, and non-custodial copy trading.
- **Why This Version Exists:** B2B enterprise partnerships with Prop Firms provide mass user acquisition channels at zero CAC, while social verification establishes market authority.

- **Features:** Prop Firm White-Label Dashboard, Public Verified Performance Links (like MyFxBook/TraderSync verified badges), Leaderboards, Copy Trading Signal Engine (Peer-to-Peer trade copying via EA bridges).
- **UI/UX Changes:** Customizable tenant branding for B2B clients (Custom logo, CSS colors, domain masking), Public Trader Profile Pages, Leaderboard discovery page, Copy Trading Allocation control interface.
- **Backend Changes:** Multi-Tenant Backend Isolation Engine, Public Verification Cryptographic Proof Generator, Master-Follower Signal Queue for Copy Engine.
- **Database Changes:** Create tenants, public_profiles, copy_relationships, signal_queue tables.
- **API Changes:** /api/v1/public/verify/{hash}, /api/v1/copy-trading/*, /api/v1/tenant/*.
- **AI Changes:** AI Bot checks for fraudulent performance manipulation before granting "Verified Leaderboard Seal".
- **Security Changes:** Strict multi-tenant data boundary isolation; Prevent public profiles from leaking ticket numbers, account IDs, or exact balance sizes (percentage-based display option).
- **Testing Requirements:** Tenant isolation unit tests (ensuring Tenant A cannot access Tenant B metrics under any vector); Signal propagation latency tests.
- **Acceptance Criteria:**
 1. B2B Prop Firm client can host Velora on journal.propfirm.com with custom branding.
 2. Public verification page proves trade authentic execution directly against server history hash.
 3. Copy trading signal replicates from master to follower EA in < 100ms.
- **Dependencies:** Version 2.0 EA & Mobile infrastructure.
- **Estimated Complexity:** Very High.
- **Development Priority:** P2 (Medium).

Version 3.0 — The Sovereign Trading OS (Voice AI, ML Models & Open API)

- **Version Goal:** Deliver the final vision of Velora: A real-time Voice-assisted AI risk co-pilot, predictive Machine Learning models, and developer Open API Marketplace.
- **Why This Version Exists:** Represents the technological pinnacle of trading journals, converting Velora from a post-trade review tool into a real-time, in-flight trading co-pilot that prevents mistakes before they happen.
- **Features:** Real-Time Voice AI Co-Pilot (WebSockets audio stream listening during active trading sessions, speaking warning prompts when rules are breached), Machine Learning Win-Probability Scoring (Predictive setup rating prior to entry), Open API Marketplace for developer extensions.
- **UI/UX Changes:** Voice Co-Pilot visualizer wave interface on desktop web; Setup Evaluator

Calculator widget; Developer API Key Management Portal.

- **Backend Changes:** WebSocket Streaming Gateway (/ws/v1/voice-copilot), Machine Learning Inference Engine Integration (Python microservice handling ML model scoring), Public Developer API Rate-Limiter Gateway.
- **Database Changes:** Create developer_api_keys, ml_model_predictions, voice_session_logs tables.
- **API Changes:** /api/v1/open/v1/*, /ws/v1/voice-copilot, /api/v1/ml/predict-setup.
- **AI Changes:** Integration of real-time speech-to-text / text-to-speech models connected to user risk profile parameters.
- **Security Changes:** OAuth2 Scoped Tokens for Developer API Marketplace; Granular rate limiting (100 req/min per API key).
- **Testing Requirements:** WebSocket connection stability tests under low bandwidth; Voice model latency measurement (< 500ms audio-to-audio return).
- **Acceptance Criteria:**
 1. Voice AI intervenes verbally within 1 second if trader opens a lot size exceeding account risk parameters.
 2. ML Win-Probability model rates trade setup based on historical context with > 65% statistical accuracy.
 3. External developers can build and monetize extensions on the Velora Open API Marketplace.
- **Dependencies:** Complete Phase 1 through Phase 3 infrastructure.
- **Estimated Complexity:** Extremely High.
- **Development Priority:** P2 (Medium).

5. Major Product Milestones

[M1: ARCHITECTURE STABLE] ----> [M2: META-API LIVE] ----> [M3: COMMERCIAL LAUNCH]		
----> [M4: EA & MOBILE LIVE] ----> [M5: SOVEREIGN AI OS]		
(Version 0.1)	(Version 0.2)	(Version 1.0)
(Version 2.0)	(Version 3.0)	
6 Weeks Completion	12 Weeks Completion	28 Weeks Completion
48 Weeks Completion	88 Weeks Completion	

6. Complete Feature Matrix Across Versions

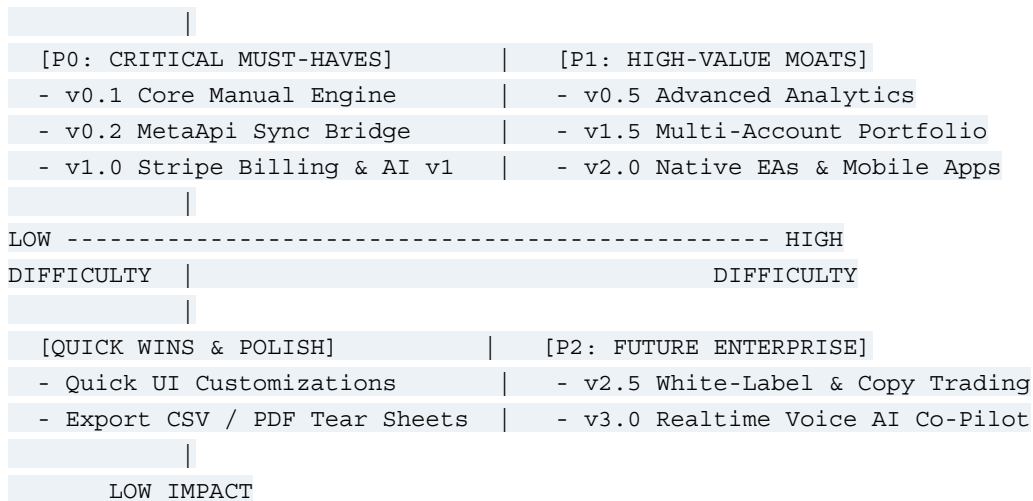
Feature	v0.1	v0.2	v0.5	v1.0	v1.5	v2.0	v2.5	v3.0
JWT Authentication & RBAC	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Manual Trade Logging	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MetaApi Auto-Sync Bridge	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Interactive Recharts Equity Engine	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Setup & Mistake Tagging	No	No	Yes	Yes	Yes	Yes	Yes	Yes

Feature	v0.1	v0.2	v0.5	v1.0	v1.5	v2.0	v2.5	v3.0
Stripe Billing & Subscriptions	No	No	No	Yes	Yes	Yes	Yes	Yes
AI Coach Insights v1 (LLM)	No	No	No	Yes	Yes	Yes	Yes	Yes
Multi-Account Aggregation	No	No	No	No	Yes	Yes	Yes	Yes
Prop Firm Rule Monitor	No	No	No	No	Yes	Yes	Yes	Yes
Custom Native MT4/ MT5 EAs	No	No	No	No	No	Yes	Yes	Yes
Native iOS &	No	No	No	No	No	Yes	Yes	Yes

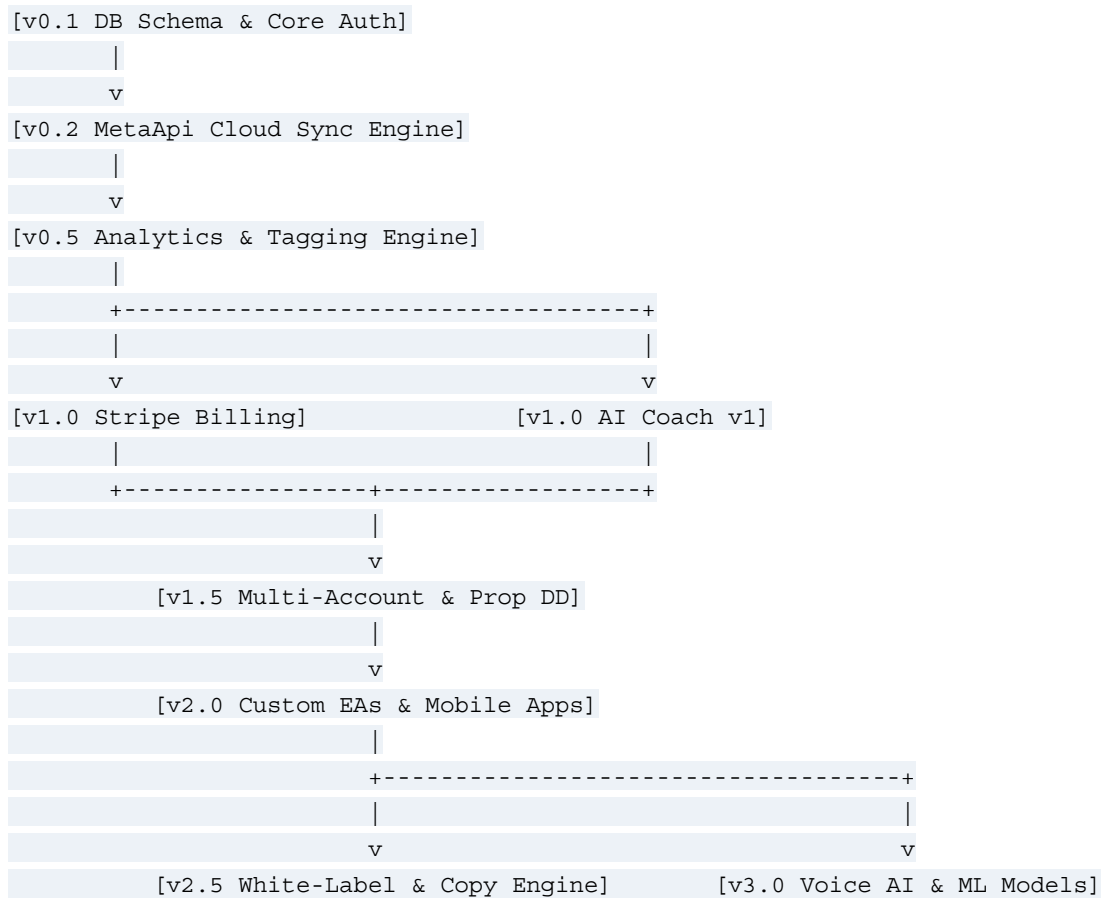
Feature	v0.1	v0.2	v0.5	v1.0	v1.5	v2.0	v2.5	v3.0
Android Apps								
White-Label Prop Firm Platform	No	No	No	No	No	No	Yes	Yes
Copy Trading Signal Engine	No	No	No	No	No	No	Yes	Yes
Voice AI Co-Pilot (Realtime)	No	No	No	No	No	No	No	Yes
Developer Open API Marketplace	No	No	No	No	No	No	No	Yes

7. Priority Matrix

HIGH IMPACT



8. Architectural Dependency Graph



9. Suggested Engineering Release Schedule

1. **Sprint Block 1 (Weeks 1–6):** Implement v0.1. Lock down DB schema, JWT security, and manual trade CRUD.
2. **Sprint Block 2 (Weeks 7–12):** Implement v0.2. Integrate MetaApi SDK, webhook listeners, and background sync queues.
3. **Sprint Block 3 (Weeks 13–20):** Implement v0.5. Build Recharts analytics components, tagging interfaces, and image upload pipelines.
4. **Sprint Block 4 (Weeks 21–28):** Implement v1.0. Connect Stripe webhooks, launch AI Coach LLM pipeline, build admin panel, and execute Commercial Launch.
5. **Sprint Block 5 (Weeks 29–36):** Implement v1.5. Build FX rate synchronization, multi-account aggregation, and Prop Firm risk monitors.
6. **Sprint Block 6 (Weeks 37–48):** Implement v2.0. Develop MQL4/MQL5 EAs and deploy React Native Mobile Apps to App Store / Google Play.
7. **Sprint Block 7 (Weeks 49–60):** Implement v2.5. Multi-tenant white-label isolation and copy-trading signal infrastructure.
8. **Sprint Block 8 (Weeks 61–88):** Implement v3.0. Voice AI WebSocket server, ML model integration, and Open API Marketplace.

10. Estimated Development Duration Summary

- **Phase 1 (MVP Foundation):** 12 Weeks (3 Months)
- **Phase 2 (Core Platform & Commercial Launch):** 16 Weeks (4 Months)
- **Phase 3 (Professional Platform & Mobile Moat):** 20 Weeks (5 Months)
- **Phase 4 (Enterprise Platform & Sovereign AI):** 40 Weeks (10 Months)
- **Total End-to-End Delivery Time (v0.1 to v3.0):** 88 Weeks (~21 Months)

11. Comprehensive Risk Analysis & Mitigations

Phase 1 Risks

- **Risk:** MetaApi service outages or account connection failures causing lost webhooks.
- **Mitigation:** Implement a fallback poll worker that sweeps all active accounts every 6 hours to pull missing historical trades via MetaApi REST endpoints.

Phase 2 Risks

- **Risk:** High LLM API costs for AI Coach under heavy user load.
- **Mitigation:** Implement aggressive prompt caching, limit AI analysis triggers to once per

week per user, and set maximum token output limits.

Phase 3 Risks

- **Risk:** MetaTrader terminal updates breaking custom MQL4/MQL5 EA compatibility.
- **Mitigation:** Build automated daily integration tests on isolated MT4/MT5 test terminals running inside Docker containers to detect EA compilation errors instantly.

Phase 4 Risks

- **Risk:** Real-time Voice AI latency exceeding human conversational thresholds (> 1.5 seconds).
- **Mitigation:** Deploy dedicated Edge WebSocket servers close to major trading hubs (London, New York, Tokyo) running quantized speech-to-text models.

12. Final CTO Operational Checklist for AI Developers

1. **No Direct DB Writes in Controllers:** Every DB action must go through a Repository class (TradeRepository.php, UserRepository.php).
2. **Strict Type Checking:** Enable declare(strict_types=1); across all PHP files and compile TypeScript with strict: true in Next.js.
3. **Database Transactions:** Always wrap multi-row mutations (e.g., trade entry + daily analytics summary update) in MySQL transactions.
4. **API Endpoint Versioning:** Prefix every route with /api/v1/ to ensure future versions do not break existing mobile or EA integrations.
5. **Zero Floating-Point Math for Currency:** Always use DECIMAL(15,2) or DECIMAL(15,5) in MySQL and string-based arbitrary precision math functions (bcadd, bcsub) in PHP to prevent currency rounding errors.